CCNA 200-301 V2.0

20 DHCP: Client, Server and Relay

Part IV — Network Services and Security

EXAM TOPICS

1.7

LECTURER

Dr. Mustafa Hameed

THE ISLAMIA UNIVERSITY OF BAHAWALPUR

Department of Information Technology

What you will be able to do

- **Describe** the four DHCPv4 messages and say which of them are broadcast and why that matters.
- **Configure** an IOS router in each of the three roles: client, server and relay agent.
- **Interpret** `show ip dhcp binding`, `show ip dhcp pool` and `show ip dhcp conflict`.
- **Troubleshoot** a host that has no address, has the wrong address, or has an address but no connectivity — and tell those three apart before touching anything.

ASSUMED KNOWLEDGE

Before we start

Chapter 4 for addressing and subnet boundaries. Chapter 8 and Chapter 10 for SVIs — a DHCP fault is very often a VLAN fault wearing a disguise. Chapter 6 for UDP and for broadcast versus unicast.

KEY TERMS

Vocabulary for this chapter

- DORA
- DHCPDISCOVER
- lease
- binding
- relay agent
- ip helper-address
- giaddr
- excluded address
- APIPA
- conflict

Each is defined where it first matters — not here.

ON THE EXAM

The verb is troubleshoot, not configure

This is what the blueprint actually asks for

Topic **1.7** reads *troubleshoot DHCPv4 client, server, and relay on IOS devices*. It is worth reading twice. The exam is not primarily asking you to build a DHCP pool from nothing — it is asking you to be shown a pool, a relay and a client that between them are not working, and to say which of the three is at fault.

Note also **DHCPv4**. There is no DHCPv6 in this topic. IPv6 hosts in this course get their addresses by SLAAC (Chapter 5).

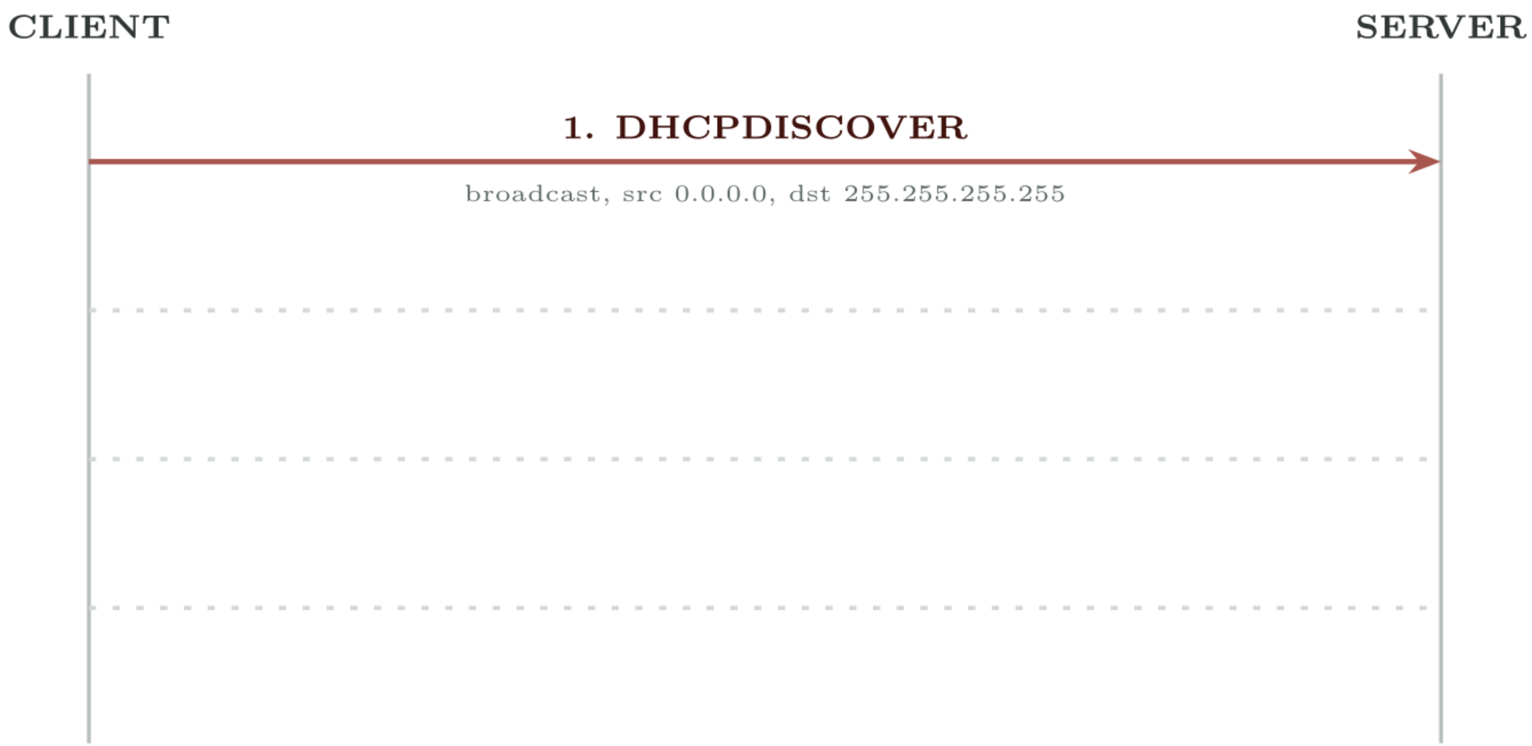
That is why this chapter spends more space on *how to tell the three failure classes apart* than on the pool syntax, which is four lines long.

DEFINITION

DHCPv4

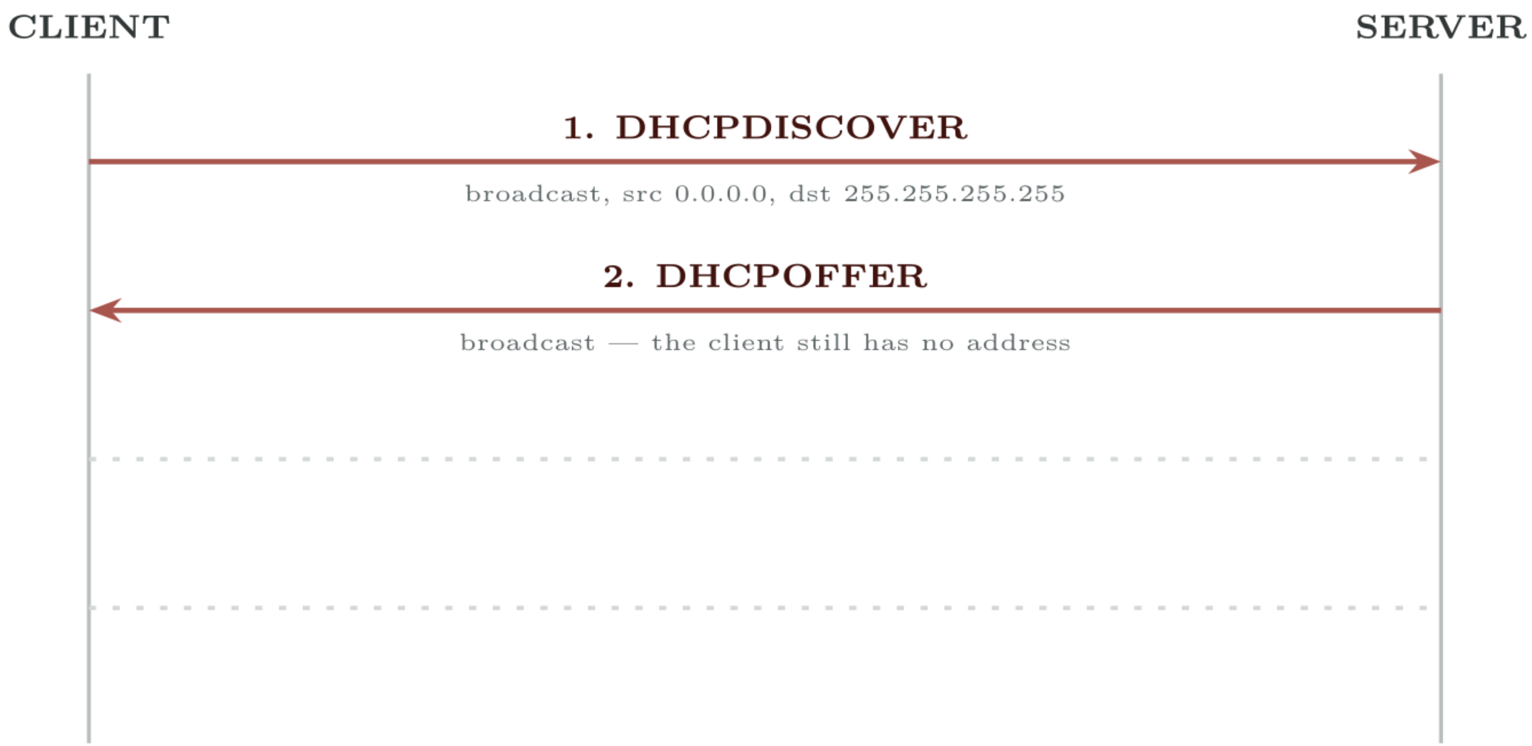
A client–server protocol that leases an IPv4 address and a set of configuration options — default gateway, DNS servers, domain name, lease time — to a host for a bounded period. The client runs on UDP port 68, the server on UDP port 67.

DORA



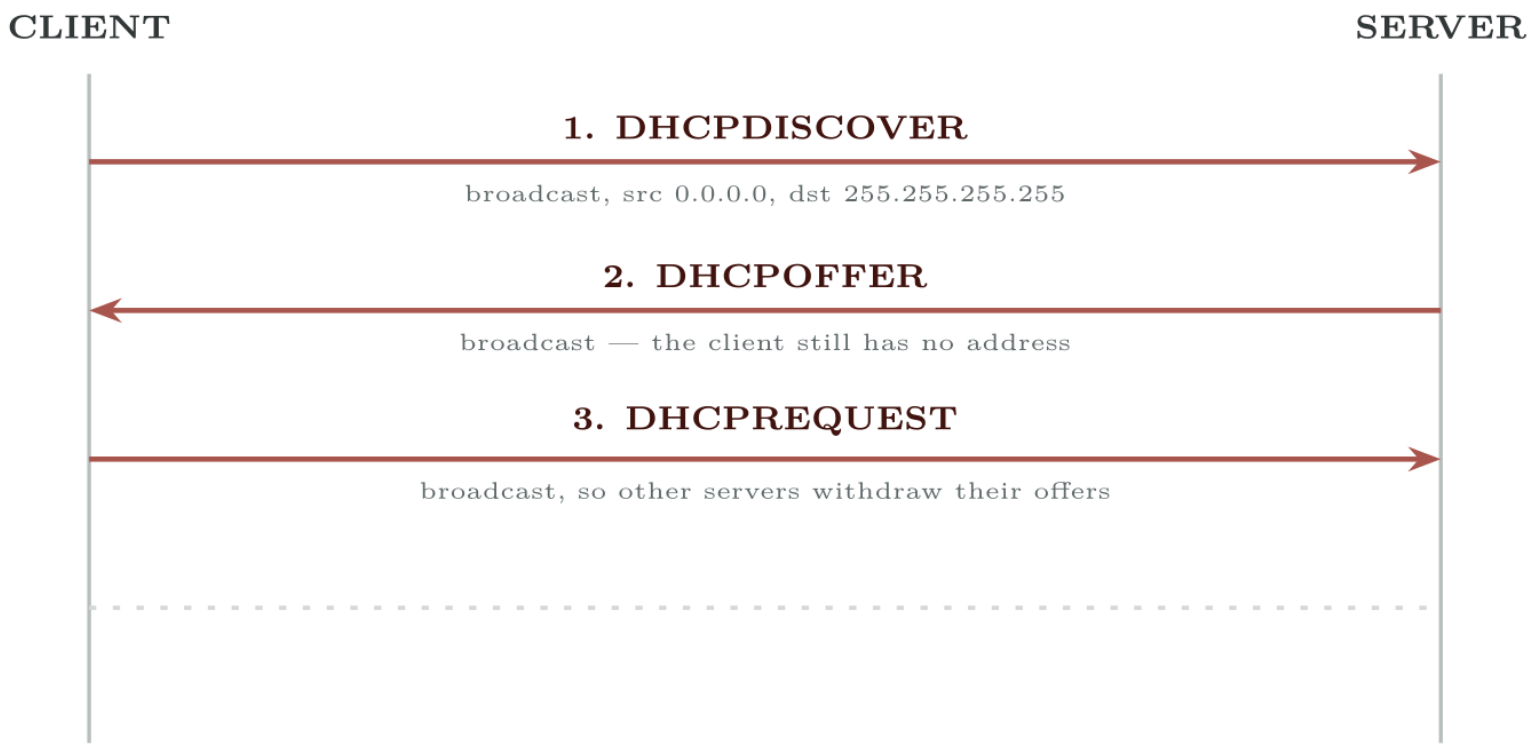
**A host with no address shouts into the broadcast domain.
Who can hear it?**

DORA



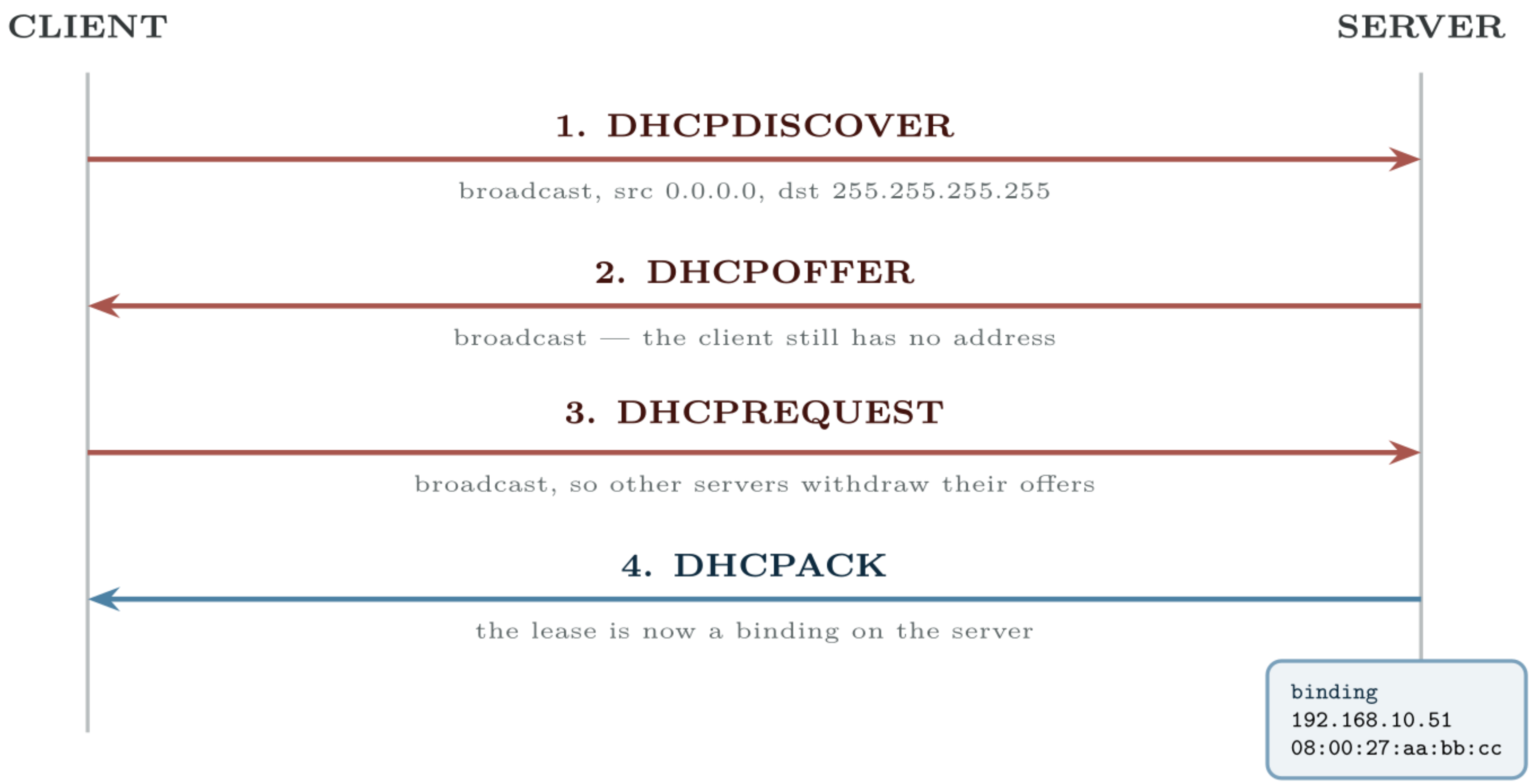
The offer is a broadcast too — the client has nothing to unicast to yet

DORA



Broadcast again, on purpose: the servers not chosen hear it and release

DORA



Only now is there a lease. Three of the four were broadcasts.

CONCEPT

Renewal is not DORA

At half the lease time (T1), a client that already has an address sends a *unicast* `DHCPREQUEST` straight to its server and expects an ACK. There is no discover and no offer. This matters for troubleshooting: a fault that only breaks broadcast forwarding — a dead relay, a wrong VLAN — can leave existing hosts working perfectly for hours while every newly booted host fails. “It works for everyone except the machines that rebooted this morning” is a relay fault until proved otherwise.

CONFIGURE

A router that leases its own WAN address

IOS · configuration mode

```
interface GigabitEthernet0/0/0
 description WAN to ISP
 ip address dhcp
 no shutdown
```

VERIFY

show ip interface brief

```
R1#

Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   203.0.113.24    YES DHCP   up                    up
GigabitEthernet0/0/1   192.168.10.1    YES manual up                    up
```

ON THE EXAM

Role 1: IOS as a client

This is what the blueprint actually asks for

A router that is a DHCP client normally also wants the default route the lease carries. It installs one automatically as a static default with an administrative distance of 254, so anything you configure yourself will win. Check with `show ip route static` if a branch router is reaching the internet by a path you did not expect.

CONFIGURE

A DHCP pool for VLAN 10

IOS · configuration mode

```
! 1. Reserve the addresses that are already spoken for -- the gateway,
!    the switch management SVIs, the printers, the servers.
!    Do this FIRST. A pool created before the exclusions can hand out
!    the gateway address in the seconds between the two commands.
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.10.250 192.168.10.254
!
ip dhcp pool VLAN10-STAFF
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.20.10 192.168.20.11
 domain-name campus.example.edu
 lease 0 8 0
!
! lease is days hours minutes. "lease 0 8 0" is eight hours, which
! suits a teaching lab; "lease infinite" suits nothing.
```

WATCH OUT

The network statement must match the interface's subnet

The router decides which pool to use by looking at the address of the interface the request arrived on — or, for a relayed request, at the `giaddr` field. If VLAN 10's SVI is `192.168.10.1/24` and the pool says `network 192.168.10.0 255.255.255.0`, they match and the pool is used.

If somebody later renumbers the SVI to `/25` and leaves the pool at `/24`, the pool no longer matches, the router silently declines to serve, and the symptom is a host with no address at all — with a perfectly valid-looking pool sitting in the configuration.

VERIFY

show ip dhcp binding

```
R1#

Bindings from all pools not associated with VRF:
IP address      Client-ID/              Lease expiration        Type      State
                Hardware address
192.168.10.21   0100.5079.6668.01       Aug 28 2026 06:14 AM    Automatic Active
192.168.10.22   0100.5079.6668.22       Aug 28 2026 06:19 AM    Automatic Active
```

VERIFY

show ip dhcp pool

```
R1#

Pool VLAN10-STAFF :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 25
 Pending event                  : none
 1 subnet is currently in the pool:
 Current index   IP address range                    Leased/Excluded/Total
 192.168.10.23   192.168.10.1     - 192.168.10.254    2  / 25 / 254
```

CONFIGURE

Relaying VLAN 10’s requests to a central server

IOS · configuration mode

```
interface Vlan10
 description Staff -- clients live here
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 10.0.0.5
```

WATCH OUT

ip helper-address goes on the interface facing the clients

This is the single most common relay mistake and it is worth stating on its own. The command belongs on the SVI or interface in the subnet where the broadcasts are — the client side. Putting it on the interface facing the server does nothing at all, because no DISCOVER ever arrives there.

If a site has six client VLANs, the command goes on six SVIs.

CONCEPT

What giaddr does, and why the server needs it

When the relay forwards the DISCOVER it fills in the **gateway IP address** field, `giaddr`, with its own address on the interface that received the broadcast — `192.168.10.1` in the example above.

That single field does two jobs. It tells the server which subnet the client is on, so the server knows which pool to draw from; and it gives the server a unicast address to send the OFFER back to, so the relay can broadcast it onward to the client.

The consequence for troubleshooting: if the central server has no pool for `192.168.10.0/24`, the relay is working perfectly and the client still gets nothing. Do not conclude “the relay is broken” from “the client got no address”.

ON THE EXAM

ip helper-address forwards more than DHCP

This is what the blueprint actually asks for

By default it relays eight UDP services, not one — among them TFTP (69), DNS (53), NetBIOS name service (137) and TACACS (49) as well as DHCP (67/68). That is occasionally useful and occasionally a surprise, and it is a fair exam question. Trim it with `ip forward-protocol udp <port>` if a site wants only DHCP relayed.

WATCH OUT

The third row is the one people waste an afternoon on

“DHCP is broken” is reported when a user cannot reach the internet. Run `ipconfig /all` on the host. If it has an address in the right subnet, DHCP has done its job perfectly and the problem is elsewhere — routing, NAT, DNS or an ACL. Check the gateway and DNS values the lease carried, then leave DHCP alone.

VERIFY

show ip dhcp conflict

```
R1#

IP address       Detection method   Detection time
192.168.10.24    Ping               Aug 27 2026 09:12 AM
192.168.10.31    Gratuitous ARP     Aug 27 2026 09:41 AM
```

Common pitfalls

- **Pool created before the exclusions.** There is a real window in which the gateway address can be leased to a laptop. Exclude first.
- **`ip helper-address` on the server-side interface.** Does nothing. It goes on the client side.
- **Pool `network` not matching the SVI's subnet or mask.** Silent refusal to serve.
- **Assuming a working host proves DHCP works.** Existing leases renew by unicast and survive faults that stop every new host dead.
- **Forgetting the SVI must be up.** An SVI with no access port in its VLAN in the up state is down, so the pool it serves is unreachable — a layer 2 fault presenting as a DHCP fault.

Common pitfalls (continued)

- **Reading `169.254.x.x` as a server problem.** It means nothing answered. Nine times in ten the fault is on the client's own switch port.

Every host in VLAN 30 gets 169.254 addresses. VLAN 10 and VLAN 20 are fine.

The DHCP server is a central router, `R1`, reached through a distribution switch. VLANs 10, 20 and 30 all have SVIs on `DSW1`, all three have pools on `R1`, and only VLAN 30 fails. Static addressing inside VLAN 30 works, and a statically addressed host in VLAN 30 can ping `R1`.

Every host in VLAN 30 gets 169.254 addresses. VLAN 10 and VLAN 20 are fine.

What would you check first — and what do you expect to see?

show run interface vlan 30

DSW1#

```
interface Vlan30
 description Lab -- clients
 ip address 192.168.30.1 255.255.255.0
!
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. There is no `ip helper-address`. VLAN 10 and VLAN 20 have one; VLAN 30 was added later and the line was missed. Everything else about VLAN 30 is correct, which is exactly why static addressing works and why the subnet is reachable — routing is fine, layer 2 is fine, and only the broadcast relay is absent.

The tell is that the fault is confined to *one* VLAN. A server fault or a routing fault would take out all three; a switch-port fault would take out one host. One whole VLAN failing while its neighbours work points at something configured per-SVI, and on this path there is only one such thing.

Restoring the relay

IOS · configuration mode

```
interface Vlan30
 ip helper-address 10.0.0.5
```

And the lesson

Fix.

Then check the second half. Confirm `R1` actually has a pool whose `network` statement matches `192.168.30.0/24`. A relay pointing at a server with no matching pool produces the identical symptom, and fixing one half of a two-part fault and declaring victory is how faults come back the next morning.

DHCP in all three roles, then broken deliberately

Topology. `SW1` carrying VLANs 10, 20 and 30; `DSW1` holding the three SVIs; `R1` as the central server on `10.0.0.0/24`; two PCs per VLAN.

1. Address the SVIs and confirm inter-VLAN routing works with static host addresses before introducing DHCP at all. Do not skip this — it separates the layer 3 work from the DHCP work.
2. On `R1`, exclude the first twenty addresses of each subnet, then build three pools with gateway, DNS and an eight-hour lease.
3. Add `ip helper-address` to VLAN 10's SVI only. Boot one PC in VLAN 10 and one in VLAN 20. Record what each receives, and explain the difference in terms of `giaddr` before adding the second helper.
4. Add the remaining helpers. Confirm all six PCs lease correctly and read `show ip dhcp binding` and `show ip dhcp pool` on `R1`.

DHCP in all three roles, then broken deliberately (continued)

1. **Break 1.** Change VLAN 20's SVI mask to `/25`, leaving the pool at `/24`. Release and renew a VLAN 20 host. Explain the result.
2. **Break 2.** Give a PC a static address inside a pool range, then force new leases until a conflict is detected. Read `show ip dhcp conflict`, fix the exclusion, and clear it.
3. **Break 3.** Remove `default-router` from VLAN 30's pool. Renew a host. It will lease successfully and reach its own VLAN only. Write down the exact test that distinguishes this from a routing failure.
4. **Extension.** Configure `R1`'s uplink as `ip address dhcp` against a second router acting as an ISP, and observe the default route that appears with distance 254.

CHECKPOINT 1 of 5

Which of the four DORA messages are broadcast, and what would break if the REQUEST were unicast?

Discuss · then advance

Checkpoint 1 of 5

Which of the four DORA messages are broadcast, and what would break if the REQUEST were unicast?

> Discover, Offer and Request are broadcast; only the Ack is unicast. If the Request were unicast, a second server that had also made an offer would never learn that its offer was declined, and would hold that address reserved for nothing until it timed out.

CHECKPOINT 2 of 5

A host has `169.254.8.31`. What single fact does that establish, and what does it *not* establish?

Discuss · then advance

ANSWER

Checkpoint 2 of 5

A host has `169.254.8.31`. What single fact does that establish, and what does it *not* establish?

> It establishes only that the host asked and **nothing answered**. It does not establish where the failure is — it could be the switch port, the VLAN, the SVI, the relay, or the server, and it says nothing about whether the server is healthy.

CHECKPOINT 3 of 5

On which interface does `ip helper-address` belong, and what does the relay write into `giaddr`?

Discuss · then advance

ANSWER

Checkpoint 3 of 5

On which interface does `ip helper-address` belong, and what does the relay write into `giaddr`?

> On the interface facing the **clients** — the SVI in the subnet where the broadcasts are. The relay writes its own address on that interface into `giaddr`, which tells the server which pool to use and gives it a unicast address to reply to.

CHECKPOINT 4 of 5

Existing hosts work; every rebooted host fails. What class of fault does that pattern point to, and why?

Discuss · then advance

ANSWER

Checkpoint 4 of 5

Existing hosts work; every rebooted host fails. What class of fault does that pattern point to, and why?

A relay or broadcast-path fault. Existing hosts renew at T1 by *unicast* directly to the server, which needs no relay, so they survive. Only newly booting hosts need the broadcast path, so only they fail.

A host has the correct address and gateway but cannot browse the web. Is this a DHCP fault? Justify the answer.

Checkpoint 5 of 5

A host has the correct address and gateway but cannot browse the web. Is this a DHCP fault? Justify the answer.

> No. DHCP delivered a correct address and a correct gateway, which is its entire job. The fault is elsewhere — routing, NAT, an ACL, or DNS. Check the DNS servers the lease carried before looking further afield.

What to take away

- DORA: Discover, Offer, Request, Ack. The first three are broadcast, which is why routers block the process by default.
- Renewal at T1 is a *unicast* request with no discover, so existing hosts survive faults that kill every new one.
- As a client: `ip address dhcp`, and the `Method` column of `show ip interface brief` proves it.
- As a server: exclude first, then a pool whose `network` statement matches the serving interface's subnet exactly.
- As a relay: `ip helper-address` on the **client-side** interface. It writes `giaddr`, which selects the pool and carries the reply back.

What to take away (continued)

- Three symptoms, three places: no address means nothing answered; wrong subnet means the wrong pool or a rogue server; right address but no connectivity is not a DHCP fault.
- Conflicts come from static addresses inside the pool range. Fix the exclusion, then `clear ip dhcp conflict *`.

Where we got to

- DORA: Discover, Offer, Request, Ack. The first three are broadcast, which is why routers block the process by default.
- Renewal at T1 is a *unicast* request with no discover, so existing hosts survive faults that kill every new one.
- As a client: `ip address dhcp`, and the `Method` column of `show ip interface brief` proves it.
- As a server: exclude first, then a pool whose `network` statement matches the serving interface's subnet exactly.

NEXT

Chapter 21 — DNS, and Diagnosing Records